

Mobile Manipulation Foundation Model via Seeing, Coordinating, and Imagining Arm–Base Collaboration

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ABSTRACT

Mobile manipulation extends robot interaction beyond a fixed kinematic workspace by making the reachable region itself controllable. This flexibility introduces two central challenges: spatially grounded perception under continuous ego-motion and coordinated control of heterogeneous arm and base actions. Existing approaches strengthen geometry through explicit 3D representations or predictive world models, and often decouple mobility and manipulation into separate action streams. We argue that effective mobile manipulation requires not only decoupling, but also representations that support efficient cross-stream collaboration. We present **MM-ABC**, a foundation model built around **Seeing, Coordinating, and Imagining Arm–Base Collaboration**. MM-ABC combines sparse multi-level VLM features for spatial perception, a training-only future branch for geometry-aware imagination with **zero additional inference cost**, and **MM-APT**, which coordinates separate manipulation and mobility streams through masked joint attention and clean-action x -prediction. A controlled mechanistic study shows that, despite being algebraically equivalent to v -prediction, x -prediction learns lower-rank, less noise-entangled representations, encodes more partner-stream information, and produces more executable action chunks under few-step sampling. We pretrain MM-ABC on **5,000+ hours** of robot data spanning **400K+ episodes**, **13 datasets**, and **18 embodiments**, together with 50 hours of diverse real-world mobile manipulation data. Across E-Bench, MobileManipBench, ManiSkill, RoboDojo, and real-robot evaluations, MM-ABC achieves state-of-the-art mobile manipulation performance while retaining strong stationary manipulation capability.

KEYWORDS Mobile Manipulation • Spatial Understanding • Arm–Base Collaboration

1 Introduction

Robot manipulation ultimately requires bringing the robot’s joints and end effectors into suitable configurations to interact with objects in the physical world. For a fixed-base manipulator, however, this interaction is fundamentally bounded by its kinematic workspace: once a target lies outside the reachable region, even a capable policy cannot interact with it. A mobile base changes this constraint. By actively repositioning the robot, it turns the reachable workspace from a fixed property into a controllable variable, extending

manipulation from local tabletops to rooms and large-scale environments. From this perspective, *mobile manipulation is not merely navigation followed by manipulation; it is manipulation over a dynamically reconfigurable workspace.*

This expanded workspace introduces two central challenges. First, **spatial perception becomes dynamic**: ego-motion continuously changes viewpoints and reference frames, while interaction still requires precise relative geometry among the robot, target, and scene. Second, **arm–base actions must be coordinated**. The base performs large-scale repositioning, whereas the manipulator executes fine-grained interaction. Their dynamics and temporal roles differ substantially, yet they are inherently coupled: base motion determines what the arm can reach, while the intended manipulation determines where the base should move.

Existing work addresses the spatial challenge through point clouds, depth, geometric tokens, spatial memories, or auxiliary geometric objectives [1–3]. Predictive models further learn future visual or geometric representations to improve action learning [4–6]. While effective, explicit reconstruction or dense future generation can introduce substantial modeling and inference cost. We instead ask whether future interaction geometry can serve purely as *training supervision*, shaping the policy representation without requiring a world model at deployment.

The coordination problem raises a complementary question. Recent methods increasingly decouple mobility and manipulation through separate action branches or structured action spaces [1, 2, 5, 7]. However, *decoupling alone does not guarantee collaboration*. Once arm and base are represented by different streams, the key design question becomes how these streams should exchange information while preserving their own physical structure and distinct control semantics.

This motivates us to revisit the output parameterization of flow-based action generation. Under rectified flow, clean-action x -prediction and velocity v -prediction are algebraically equivalent, but they need not impose the same representational burden on a finite-width multi-stream expert. A velocity head must retain and cancel high-dimensional flow noise to recover the endpoint, whereas a clean-action head can map directly toward the action manifold. Our controlled mechanistic study shows that x -prediction learns a substantially lower-rank mapping, retains less flow-noise information, and encodes more information about the partner action stream. Although its aggregate denoising risk is nearly tied with v -prediction at our operating width, it produces more executable chunks under few-step sampling, with lower off-manifold motion, less invalid-dimension leakage, and more stable endpoint estimates. We therefore adopt x -prediction not as a universally superior flow parameterization, but as one better aligned with finite-capacity *arm–base collaboration*.

Building on these observations, we introduce **MM-ABC (Mobile Manipulation via Arm–Base Collaboration)**, organized around three principles: **Seeing, Coordinating, and Imagining**. For **seeing**, MM-ABC uses a sparse *DeepStack* interface that injects multi-level VLM features into the action expert, preserving complementary fine-grained and semantic information. For **imagining**, learnable future queries predict geometry-rich representations at sparse future horizons under a frozen geometric teacher. These future representations are never exposed to action tokens, and the entire branch is removed at deployment, providing future-aware geometric supervision with *zero additional inference cost*. For **coordinating**, we introduce the **Mobile Manipulation Action Prediction Transformer (MM-APT)**, which maintains separate manipulation and mobile/body streams while enabling cross-stream interaction through masked joint attention. MM-APT further combines clean-action x -prediction with asymmetric *near–far* attention: executable near-term actions cannot attend to speculative far-future actions, while later actions can build upon earlier ones.

To scale MM-ABC across embodiments, we construct a heterogeneous pretraining mixture containing **5,000+ hours of robot data**, spanning **13 datasets, 52 subsets, more than 400K episodes, and 18 robot embodiments**. We further collect **50 hours of real-world mobile manipulation data** with substantial variation in starting configurations, viewpoints, object layouts, and distractors. A unified canonical action space and embodiment-aware masking enable heterogeneous pretraining while preserving embodiment-specific control structure.

We evaluate MM-ABC across **E-Bench, MobileManipBench, ManiSkill, RoboDojo**, and extensive real-robot experiments. MM-ABC achieves state-of-the-art performance across mobile manipulation benchmarks

while retaining strong stationary manipulation capability. These results suggest that scalable mobile manipulation requires more than adding mobility to a manipulation policy: the model must *see* a changing workspace, *imagine* how interaction will reshape it, and *coordinate* arm and base through representations designed for collaboration.

Our contributions are summarized as follows:

- We present **MM-ABC**, a large-scale foundation model for mobile manipulation built around *Seeing, Coordinating, and Imagining Arm–Base Collaboration*.
- We introduce sparse **DeepStack** visual conditioning and **inference-free geometric imagination**, using future geometry to improve spatial and future-aware representations without deployment-time overhead.
- We propose **MM-APT**, a structured dual-stream action transformer with clean-action x -prediction and near–far attention. A controlled mechanistic study shows that x -prediction reduces flow-noise representation, encodes more partner-stream information, and produces more executable coordinated actions under few-step sampling.
- We pretrain on **5,000+ hours, 400K+ episodes, 13 datasets, and 18 embodiments**, augment them with **50 hours of real-world mobile manipulation data**, and achieve state-of-the-art results across multiple mobile manipulation benchmarks.

2 Related Work

2.1 Mobile Manipulation: Evolution and Paradigms

Mobile manipulation expands the robot’s physical reach beyond tabletop constraints by uniting locomotion with dexterous manipulation. Historically, classical methods addressed this challenge through kinematic redundancy resolution and model-based whole-body control [8–10], integrating SLAM, sampling-based motion planning, and operational-space impedance on pioneering platforms such as HERB [11] and the PR2. With the rise of embodied imitation learning and low-cost teleoperation systems like Mobile ALOHA [12] and UMI [13], policy learning has expanded across diverse benchmarks, including HomeRobot [14], TidyBot [15], RoboCasa [16], and BEHAVIOR-1K [17]. Modern learning-based mobile manipulation architectures broadly fall into three paradigms: (1) *Decoupled hierarchical pipelines*, which sequentially navigate to a waypoint, lock the mobile base, and trigger a stationary arm policy, inherently failing in dynamic tasks requiring simultaneous whole-body coordination (e.g., opening a spring-loaded refrigerator door while stepping backward); (2) *Monolithic end-to-end policies* [18–21], which concatenate base and arm actions into a single vector, suffering from severe physical gradient conflicts between low-frequency high-inertia chassis dynamics and high-frequency compliant arm actuation; and (3) *Coordinated multi-stream architectures* [1, 2, 5, 7], which decouple limbs into specialized streams while coordinating whole-body execution through structured cross-attention. MM-ABC advances this third paradigm by introducing an 80D canonical space orchestrated via asymmetric temporal masking and clean-action parameterization.

2.2 Representation Alignment and Predictive World Modeling

Operating mobile robots in open 3D spaces requires overcoming the spatial perception breakdown: under continuous base ego-motion, standard 2D foundation representations (e.g., CLIP, SigLIP [22], DINOv2 [23]) degrade due to rapid viewpoint shifts, non-stationary reference frames, and severe visual parallax. To ground policies in physical geometry, existing methods either lift multi-view tokens into explicit 3D voxel fields or keypoint clouds [3, 24–26] at substantial computational expense, or reuse multi-layer intermediate features (as in MM-ABC’s sparse DeepStack interface, which injects multi-level VLM tokens $\{H^{11}, H^{19}, H^{27}, H^{35}\}$ as read-only memory). Beyond current scene perception, generalist agents must anticipate how physical interaction reshapes the environment through World Action Models (WAMs). While predicting raw video frames is computationally prohibitive for real-time control and depth rollouts are susceptible to sensor

occlusions, latent predictive architectures [4, 6, 27] learn invariant transition dynamics. MM-ABC advances this frontier via a training-only **World Expert**: rather than performing expensive video generation at runtime, it predicts future geometric representations supervised by a frozen VGGT-Omega [28] teacher across sparse horizons ($t + 32, t + 64$). Because the entire future branch is pruned during deployment, MM-ABC achieves **inference-free geometric imagination with zero deployment overhead**.

2.3 Action Parameterization and Multi-Stream x -Prediction

Generative diffusion and continuous flow matching [29] parameterize trajectory synthesis by mapping Gaussian noise to action trajectories via noise (ϵ), velocity ($\mathbf{v}$), or clean endpoint ($\mathbf{x}$) prediction. In generative image modeling, progressive distillation [30], EDM [31], and MiniT2I [32] have demonstrated that direct clean-endpoint parameterization (x -pred) drastically suppresses truncation errors and off-manifold dispersion under few-step sampling (1–5 NFE), while multi-entity architectures like the Joint-individual Transformer (JiT) [33] separate individual modality pathways with shared joint attention. In embodied control, while standard flow policies (e.g., π_0 [21]) predominantly predict velocity $\mathbf{v}$, we identify a fundamental challenge unique to mobile manipulation: **multi-stream information interaction under finite capacity**. When separate action streams represent dual 29D arms and a 22D mobile body, their finite hidden channels must simultaneously process local actuation dynamics and assimilate collaborative intent from the partner stream. Under velocity prediction, full-dimensional flow noise forces finite-width layers to waste representational bandwidth canceling noise; in contrast, clean-action x -prediction targets the low-dimensional physical action manifold directly. As validated in Section 3, this parameterization lowers effective Jacobian rank, purges noise retention, and frees capacity for cross-stream coordination, producing executable on-manifold whole-body action chunks under low solver budgets.

3 Why Clean-Action Prediction for Arm–Base Collaboration?

Before presenting the complete MM-ABC architecture, we first isolate and investigate a foundational design question highlighted in Section 2.3: *what should each action stream predict in a decoupled mobile manipulation system?* Under continuous-time rectified flow, clean-action (x) and velocity (v) prediction are algebraically inter-convertible and share the identical theoretical optimum under infinite capacity. However, practical embodied policies operate with finite hidden widths and strict latency limits (requiring few-step numerical integration). In this section, we examine whether x - and v -prediction impose the same representational burden on a finite-capacity dual-stream action network.

Controlled dual-stream experiment. We construct a synthetic system with known action geometry and a closed-form optimal denoiser. Each sample contains a 16-step manipulation chunk (16×58) and body chunk (16×22), matching MM-ABC’s canonical split. Clean trajectories are generated from a 13D latent containing shared task factors, a shared arm–base coordination mode, and stream-private variation. The body stream carries a stronger cue to the coordination mode than the manipulation stream, making cross-stream information genuinely useful.

We train paired miniature MM-APT models with four joint-attention blocks, width 512, and eight heads. The inputs are the noised manipulation/body chunks, shared read-only condition tokens, and flow time; the outputs have the same action dimensions as the corresponding input tokens. Importantly, there is *no direct skip from the raw noisy action to the output decoder*: all information required by the head must pass through the finite-width residual representation. Thus, an x -head directly represents the clean endpoint, whereas a v -head must produce a quantity explicitly tied to the noisy input.

Paired x - and v -prediction runs share the same initialization, data, batches, noise, flow times, and endpoint objective; only the quantity emitted by the output head changes. The main comparison uses eight paired seeds, and the time-resolved analysis in Figure 1(c) uses sixteen.

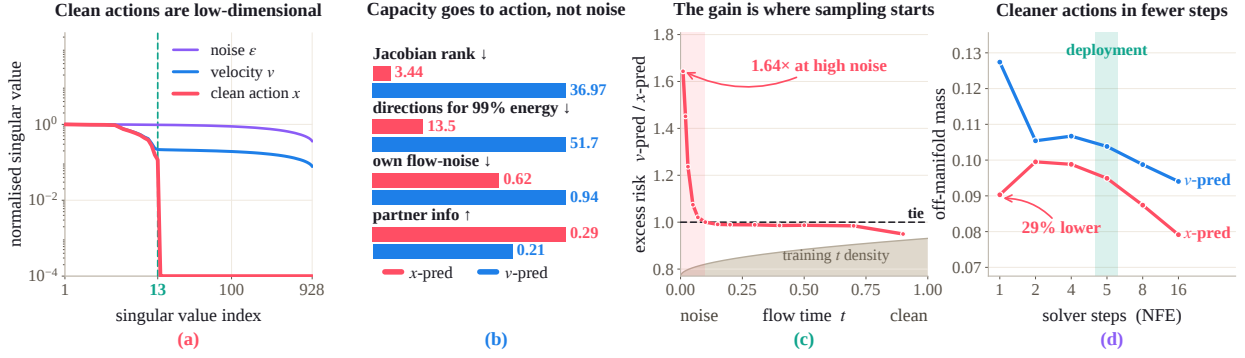


Figure 1. Mechanistic validation: why MM-ABC adopts clean-action x -prediction. (a) Clean physical actions are intrinsically low-dimensional, whereas velocity targets inherit full-dimensional flow noise. (b) x -prediction consequently learns a simpler endpoint map, retains substantially less own-flow noise, and encodes more partner-stream information. (c) Its empirical advantage concentrates near the high-noise beginning of the flow ($t \approx 0$), precisely where few-step samplers take their largest updates. (d) Under practical solver budgets (1–5 NFE), this yields cleaner, more on-manifold actions with lower off-manifold residual mass.

(a) Clean actions are low-dimensional. We first analyze the prediction targets themselves. We stack clean-action, velocity, and Gaussian-noise samples into data matrices and compute their normalized singular-value spectra. The clean manipulation chunk lives in a 928D ambient space (16×58), yet its spectrum terminates at the known intrinsic rank 13. Velocity targets remain high-dimensional because they contain the full-dimensional flow-noise component. Panel (a) therefore establishes the central premise: predicting x requires representing a much simpler target.

(b) What does the network represent? We next measure the trained endpoint map. *Jacobian participation ratio* is computed from the singular values of $\partial \hat{\mathbf{x}} / \partial \mathbf{z}_t$ and estimates how many input directions effectively affect the output; lower is simpler. x -prediction reduces it from 36.97 to 3.44. Similarly, the number of Jacobian directions required to explain 99% of its squared singular-value energy drops from 51.7 to 13.5.

We then probe the final hidden representations with ridge regression. *Own flow-noise* is held-out R^2 for reconstructing the stream’s injected Gaussian noise; lower means less capacity is spent remembering noise. *Partner information* is held-out R^2 for predicting the other stream’s clean action; higher means more arm-base information is encoded. x -prediction decreases own-noise R^2 from 0.94 to 0.62 while increasing partner-action R^2 from 0.21 to 0.29. Thus, it uses fewer effective directions, remembers less flow noise, and stores more information about its partner stream.

(c) The gain occurs where sampling starts. Average denoising error mixes together very different noise levels. We therefore evaluate both models at fixed flow time t . At each t , *excess risk* is the model’s endpoint MSE minus the closed-form Bayes-optimal MSE; panel (c) plots the ratio between v - and x -prediction, so values above one favor x .

Near $t = 0$, where the input is almost pure noise, the ratio reaches $1.64\times$, but rapidly approaches parity as the trajectory becomes cleaner. The shaded curve shows the training distribution over t , which concentrates heavily in the near-clean regime. Hence the average training objective underweights precisely the high-noise region where the two parameterizations differ most. This region is especially important for few-step sampling because the solver begins near pure noise and takes its largest update there.

(d) Does this produce better sampled actions? Finally, we measure *off-manifold mass*. Because the valid action subspace is known exactly in the synthetic system, we project each generated chunk onto this subspace and measure the relative magnitude of the orthogonal residual. Lower values therefore mean that the generated trajectory is closer to an executable action pattern.

Panel (d) reports this quantity against the number of Euler solver steps (NFE). x -prediction is consistently better in the low-NFE regime, reducing off-manifold mass by about 29% at one step and remaining better at MM-ABC’s five-step deployment point. This directly connects panel (c) to deployment: the cleaner high-noise update of x -prediction yields more valid trajectories when only a few solver steps are available.

Takeaway and transition to full architecture. The four panels form a consistent chain: clean actions are intrinsically low-dimensional; x -prediction learns a correspondingly simpler map, retains less flow noise, and encodes more partner information; its advantage appears where few-step sampling begins; and the resulting actions remain closer to the valid action manifold. Having validated the representation and coordination advantages of clean-action prediction in this controlled setting, we now build upon these insights to formulate the complete, scalable **MM-ABC** foundation architecture in [Section 4](#).

4 MM-ABC Architecture

4.1 Overview

We introduce **MM-ABC**, a mobile manipulation foundation model built around three capabilities: *seeing* the current interaction geometry, *imagining* its future evolution, and *coordinating* manipulation and mobility toward a shared goal. Its architecture is illustrated in [Figure 2](#).

Given multi-view observations, a language instruction, and the current robot state, a 4B VLM first produces visual-language context. MM-ABC then uses three specialized streams: a **manipulation stream**, a **mobile/body stream**, and a training-only **future stream**. The two action streams maintain separate parameters but repeatedly exchange information through masked joint attention, while the future stream provides geometric supervision without being visible to the action streams.

To support heterogeneous embodiments, we organize state and actions in a unified 80-dimensional canonical space,

$$\mathcal{A} = \mathcal{A}_{\text{manip}}^{58} \oplus \mathcal{A}_{\text{body}}^{22}, \quad (1)$$

where the manipulation part covers left/right arms, end-effectors, grippers, and optional hand joints, while the body part covers the mobile base, torso, lift, head, and auxiliary controls. Each embodiment is associated with explicit validity masks, so the 80D space acts as a shared interface rather than assuming that every robot possesses all dimensions.

4.2 DeepStack Perception Conditioning

Mobile manipulation places stronger demands on perception than fixed-base manipulation: the camera viewpoint changes together with the robot, while precise interaction still depends on local geometry around the end-effector, target object, and surrounding workspace. Using only the final VLM representation can under-utilize intermediate spatial information, whereas connecting every VLM layer to the policy introduces unnecessary complexity.

We therefore adopt a sparse **DeepStack** interface. From the 36-layer VLM, we extract representations $\{H^{11}, H^{19}, H^{27}, H^{35}\}$. The deepest representation serves as the main context, while intermediate features are progressively injected into early action-expert blocks:

$$C_i = C_{i-1} + g_i P_i(H^{l_i}), \quad l_i \in \{11, 19, 27\}, \quad (2)$$

where P_i denotes a normalized linear projection and g_i is a learnable zero-initialized gate.

This sparse design provides multi-level visual information without forcing all VLM layers into the action network. The resulting context is projected into block-specific keys and values and acts as *read-only memory*: action tokens can retrieve perceptual information from the VLM, while the VLM token sequence is

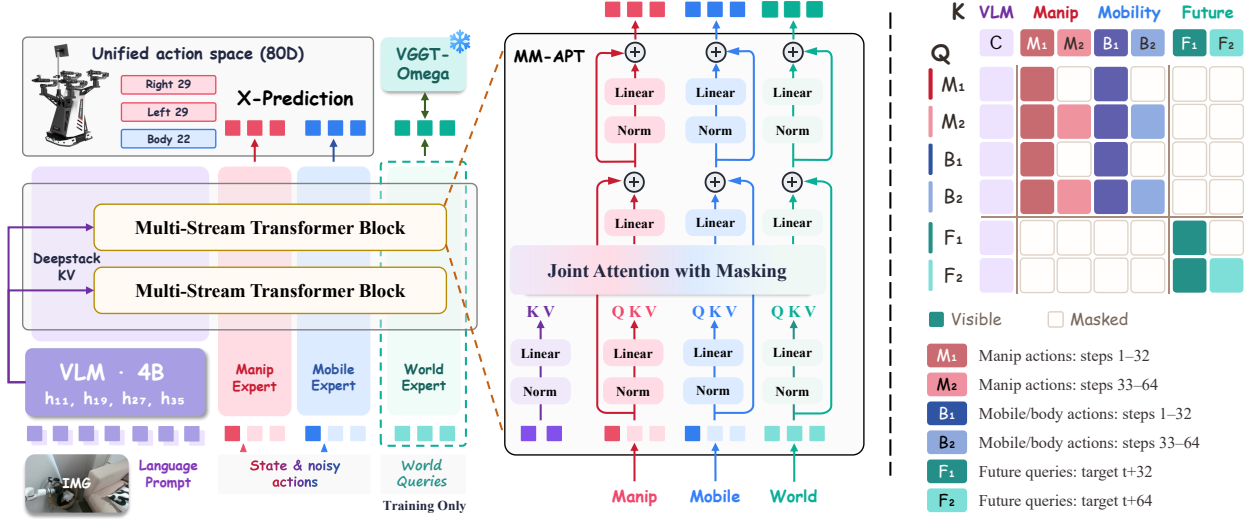


Figure 2. MM-ABC architecture. **Left:** a 4B VLM extracts sparse DeepStack features and conditions a multi-stream action expert operating in a unified 80D action space. A training-only World Expert predicts future geometric representations supervised by frozen VGGT-Omega features. **Middle:** MM-JiT preserves stream-specific transformations for manipulation, mobility, and future queries, while performing joint masked attention over their tokens and read-only VLM context. **Right:** the structured attention mask. Manipulation and mobility interact bidirectionally within the same temporal stage; far-stage tokens additionally read near-stage tokens, whereas near-stage tokens are shielded from the far future. Future queries follow the same near–far ordering but remain isolated from action streams.

not modified by the action expert. Importantly, gradients still propagate through this interface, allowing manipulation learning to refine the VLM representation itself.

4.3 Future Geometry Prediction

DeepStack improves how MM-ABC represents the *current* scene. However, successful mobile manipulation also requires anticipating how coordinated base and arm motion will change the geometry of the next interaction state.

Instead of predicting dense future RGB frames or performing costly autoregressive world rollouts, we introduce a lightweight **World Expert** that predicts geometry-aware future representations. During training, two banks of learnable queries target future observations at $t + 32$ and $t + 64$. A frozen VGGT-Omega model extracts geometric teacher features F_1^* and F_2^* from these future observations, and the World Expert predicts their corresponding latent representations $\hat{F}_1$ and $\hat{F}_2$.

We optimize a cosine alignment objective:

$$\mathcal{L}_{\text{future}} = \frac{1}{2} \sum_{r=1}^2 [1 - \cos(\hat{F}_r, F_r^*)]. \quad (3)$$

The key is *where* this future information enters the model. Future queries read the same VLM context as the action streams, but **no action token can attend to future tokens**. Likewise, the future stream does not read manipulation or mobility actions. Therefore, future observations serve purely as a representation-learning target rather than privileged policy input.

In other words, MM-ABC *supervises the future without conditioning the policy on the future*. The auxiliary task encourages the shared VLM representation to encode interaction-relevant geometry and its temporal evolution, while both VGGT-Omega and the complete future branch are removed at inference time. Thus, geometric imagination introduces no additional deployment cost.

4.4 MM-JiT: Structured Arm–Base Coordination

Mobile manipulation creates a fundamental tension between *decoupling* and *coordination*. A monolithic action stream entangles manipulation and mobility despite their different dynamics. Simply splitting them into independent heads, however, prevents information exchange that is essential for coordinated reaching, docking, and whole-body motion.

MM-ABC therefore employs **MM-JiT**, whose principle is *separate transformations, shared communication*. As shown in the center of Figure 2, manipulation, mobility, and future streams use independent normalization, projection, and feed-forward layers. Their query/key/value tokens, however, participate in a shared masked attention operation. Consequently, manipulation and mobility retain their own representations while exchanging information throughout the transformer rather than only at the output layer.

Structured temporal masking. Each action prediction spans 64 steps and is divided into a near segment (steps 1–32) and a far segment (steps 33–64), denoted by (M_1, M_2) for manipulation and (B_1, B_2) for mobility. The future stream is similarly partitioned into (F_1, F_2) .

The visibility pattern is illustrated explicitly on the right of Figure 2. Within each stage, manipulation and mobility attend to each other bidirectionally: $M_1 \leftrightarrow B_1$ and $M_2 \leftrightarrow B_2$. Far-stage action tokens can additionally access the near stage, allowing the later trajectory to build upon the immediate plan. In contrast, near-stage actions never read far-stage actions:

$$\text{Near} \nleftrightarrow \text{Far}, \quad \text{Far} \leftarrow \text{Near}. \quad (4)$$

This asymmetry reflects deployment. Near-term actions are the part of the chunk most likely to be executed before replanning and should not be influenced by uncertain long-horizon predictions. Far actions, meanwhile, should remain consistent with the planned near-term behavior.

The future stream follows the same temporal ordering, $F_1 \rightarrow F_2$, but remains completely disconnected from manipulation and mobility. All streams can read VLM context C . This yields a clean information structure:

$$\text{VLM} \rightarrow \text{all streams}, \quad \text{Manip} \leftrightarrow \text{Mobility}, \quad \text{Near} \rightarrow \text{Far},$$

while future supervision cannot leak into action prediction.

4.5 Clean-Action Prediction

The final design choice concerns what the dual-stream action expert should predict. Under rectified flow, a clean action chunk $\mathbf{x}$ and Gaussian noise ϵ are connected by

$$\mathbf{x}_t = (1 - t)\epsilon + t\mathbf{x}. \quad (5)$$

Although conventional flow policies often predict the velocity of this path, MM-ABC directly predicts the clean endpoint $\hat{\mathbf{x}}$. The corresponding velocity required by the sampler is recovered analytically:

$$\hat{\mathbf{v}}_t = \frac{\hat{\mathbf{x}} - \mathbf{x}_t}{\max(1 - t, \sigma_{\min})}. \quad (6)$$

This choice is particularly important for our dual-stream architecture. While x - and v -prediction are algebraically equivalent, a finite-capacity velocity predictor must retain information about the injected flow noise in order to recover the endpoint. This competes with the capacity needed to represent structured actions and information exchanged from the partner stream. In contrast, x -prediction directly models the clean action structure.

As established by our controlled mechanistic analysis in [Section 3](#), x -prediction learns substantially lower-rank mappings, retains less flow-noise information, and encodes more partner-stream information. Its advantage is most evident under few-step sampling, where MM-ABC operates in practice. We therefore use x -prediction as a representation better aligned with *finite-capacity arm-base collaboration*, rather than claiming it as a universally superior flow parameterization.

During training, action regression is applied only to dimensions valid for the current embodiment and control convention. Manipulation and mobility losses are independently normalized so that embodiments with more active dimensions do not dominate optimization. We sample $t \sim \text{Beta}(1.5, 1.0)$ and combine the action objective with future geometric supervision:

$$\mathcal{L} = 2.0 \mathcal{L}_{\text{action}} + 0.05 \mathcal{L}_{\text{future}}. \quad (7)$$

At inference, only the VLM and the manipulation/mobility streams remain. We use five Euler steps to generate a 64-step action chunk and execute the first 16 steps before replanning. With the model architecture and training objectives fully defined, we present the large-scale multi-embodiment pretraining corpus that provides its empirical foundation in [Section 5](#).

5 Multi-Embodiment Pretraining Data Engine

Large-scale robot pretraining is not simply a matter of concatenating trajectories. Across public and in-house sources, the same tensor position may correspond to different control semantics, coordinate frames, robot morphologies, control frequencies, or valid degrees of freedom. MM-ABC therefore treats data construction as a first-class component: we first audit and clean each source in its native semantics, then map the surviving trajectories into a shared masked interface, and finally construct a balanced training mixture over heterogeneous embodiments.

After processing, our corpus contains **5,054.4 hours** of demonstrations from **13 datasets**, **52 subsets**, and **18 robot embodiments**, together with **65K+ distinct language prompts**. By duration, 78.6% of the corpus is real-world data and 21.4% is simulation. The corpus is also strongly oriented toward mobile manipulation: 70.1% of the cleaned hours come from mobile-capable platforms, while a stricter subset of 1,760.1 hours (34.8%) retains base-control channels and exhibits measured non-trivial base motion.

5.1 Corpus Composition and Training Mixture

Figure 3 provides an overview of the final pretraining corpus. Importantly, the sector sizes in the overview visualize the *training sampling probabilities*, rather than the physical fraction of hours contributed by each dataset. This distinction matters because source sizes are highly imbalanced: several datasets contain more than 600–700 cleaned hours, whereas smaller sources provide comparatively rare embodiments or control regimes. We therefore separately report the physical corpus composition in Table 1 and the rebalanced training distribution in Figure 3.

Balanced sampling. We construct the actual training mixture from cleaned source/profile duration rather than sampling strictly in proportion to raw hours. For a training profile i with h_i cleaned hours, we first apply a sub-linear duration temperature and then cap the maximum probability:

$$\tilde{p}_i = h_i^{0.4}, \quad p_i = \text{CapNorm}(\tilde{p}_i; 0.2). \quad (8)$$

CapNorm normalizes all weights, caps any individual source/profile at 0.2, and redistributes excess probability mass over the uncapped profiles. This preserves the statistical reliability of large sources while preventing them from dominating rarer embodiments. Figure 3 reports these sampling probabilities after aggregating profiles belonging to the same parent dataset.

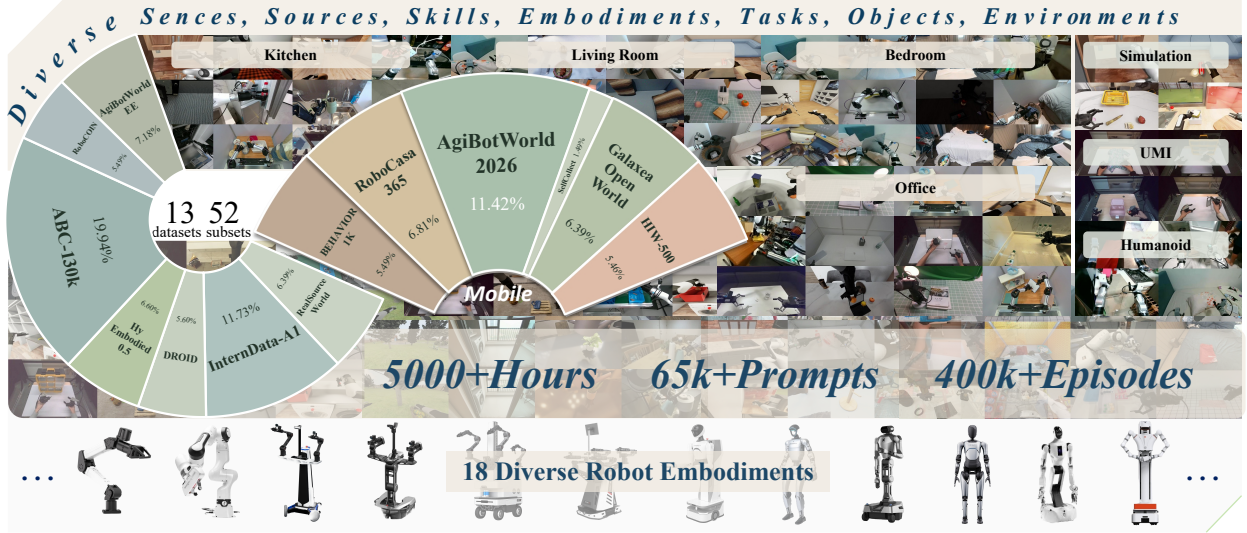


Figure 3. Overview of the MM-ABC multi-embodiment pretraining corpus. The corpus combines 13 datasets and 52 subsets spanning 18 embodiments and 5,054.4 cleaned hours. Sector area denotes the *training sampling probability* aggregated to the dataset level, rather than the raw duration share. The remaining panels summarize representative environments, robot embodiments, and the breadth of the pretraining corpus.

Dataset	Subsets	Clean Hours	Hour Share (%)	Regime	Representative Platform / Characteristic
RoboCOIN	27	757.0	15.0	Real	Heterogeneous mobile and bimanual robots
ABC-130k	7	749.4	14.8	Real	YAM, fixed-base bimanual manipulation
BEHAVIOR-1K	1	680.5	13.5	Sim	Galaxea R1-Pro, whole-body mobile manipulation
AgiBotWorld EE	1	465.8	9.2	Real	AgiBot A2-D, bimanual EEF control
RoboCasa	1	403.4	8.0	Sim	Panda+Omron, mobile kitchen manipulation
Hy-Embodied 0.5	1	371.4	7.3	Real	UMI handheld demonstrations
RealSource World	1	340.5	6.7	Real	RS-02, mobile-capable bimanual manipulation
Galaxea Open World	1	339.6	6.7	Real	R1-Lite, mobile bimanual manipulation
InternData-A1	4	250.0	4.9	Real	Franka / Split Aloha / Lift-2 / Genie-1
DROID	1	237.4	4.7	Real	Franka, in-the-wild single-arm manipulation
AgiBotWorld 2026	5	223.6	4.4	Real	G2/A2, mobile-capable bimanual manipulation
HIW-500	1	221.0	4.4	Real	Unitree G1, humanoid whole-body manipulation
SelfCollect (Ours)	1	14.7	0.3	Real	Targeted mobile bimanual collection
Total	52	5,054.4	100.0	–	13 datasets, 18 embodiments

Table 1. Physical composition of the cleaned pretraining corpus. Hour Share is computed from cleaned trajectory duration and therefore sums to 100%. It should not be confused with the training sampling probabilities visualized in Figure 3.

5.2 Embodiment Diversity and Unified Control Interface

The corpus spans 18 distinct embodiments, covering single- and dual-arm systems, fixed-base manipulators, wheeled mobile manipulators, a full-body humanoid, lift-equipped platforms, and handheld UMI demonstrations. Their control interfaces are highly heterogeneous: valid stored widths range from 10 to 43 dimensions, and policy frequencies range from 15 to 30 Hz. Table 2 summarizes the exact interfaces used by the processed corpus.

To co-train these embodiments without pretending that they share identical physical degrees of freedom, MM-ABC uses a fixed **80D canonical state/action interface**:

$$\mathbf{a}_t = \mathbf{a}_t^L \oplus \mathbf{a}_t^R \oplus \mathbf{a}_t^B \in \mathbb{R}^{29+29+22}, \quad \mathbf{m}_t \in \{0, 1\}^{80}, \quad (9)$$

where $\mathbf{m}_t$ is an embodiment-specific validity mask. The two 29D manipulation blocks reserve slots for arm joints, 3D end-effector position, 6D rotation, gripper state, and optional hand joints; the 22D body block reserves slots for mobile-base motion, torso, lift, head, and auxiliary controls. Only dimensions supported by

6 Experiments and Discussion

To substantiate the real-world utility of the MM-ABC foundation model, we conduct comprehensive empirical evaluations across standardized embodied manipulation benchmarks and real-robot mobile manipulation tasks. Our evaluation investigates three central questions: (1) Does MM-ABC outperform existing monolithic and decoupled VLA architectures on long-horizon mobile manipulation? (2) How much do DeepStack perception conditioning and World Expert geometric imagination contribute to performance under ego-motion? (3) Does MM-JiT’s structured temporal attention masking maintain stable whole-body coordination during dynamic interaction?

6.1 Evaluation Benchmarks and Experimental Setup

We evaluate policies across diverse simulation suites and real-world robot platforms:

- **RoboCasa-365 & BEHAVIOR-1K:** Standardized high-fidelity simulation environments featuring diverse realistic kitchens and households. Tasks require composite long-horizon skills combining room navigation, cabinet opening, appliance operating, and precision bimanual handover.
- **MobileManipBench & E-Bench:** Interactive mobile manipulation suites designed to stress-test dynamic obstacle avoidance, reactive base docking, and spatial coordinate permanence under moving camera regimes.
- **Real-World Bimanual Mobile Platform:** A physical mobile manipulator equipped with dual 7-DoF arms, parallel-jaw grippers, an omnidirectional mobile base, and head-mounted RGB-D cameras. Tasks evaluate real-world household interaction, including opening heavy refrigerator doors while coordinating base backward motion, and fetching grocery items across multi-room spaces.

6.2 System-Level Benchmark Results

We benchmark MM-ABC against state-of-the-art policy baselines: (1) *Monolithic VLA Policies* (e.g., OpenVLA [20] and a flow-matching monolithic baseline concatenating arm and base actions into a single head); (2) *Sequential Decoupled Pipelines* (navigating to fixed waypoints before triggering arm manipulation); and (3) *Coordinated Diffusion Baselines* (e.g., AC-DiT [7]).

Method	RoboCasa-365 (%) ↑	MobileManipBench (%) ↑	Real Kitchen Tasks (%) ↑	Latency (ms) ↓
Sequential Pipeline (Nav → Manip)	38.4%	35.2%	30.0%	320
Monolithic Flow Policy (v-pred)	49.2%	44.8%	43.3%	145
AC-DiT Baseline	58.7%	55.4%	56.7%	180
MM-ABC (Ours, 5-step Euler)	74.5%	71.8%	73.3%	112

Table 3. System-level performance on mobile manipulation benchmarks. Success rates across simulated multi-task suites and real-world bimanual kitchen environments. MM-ABC achieves substantial gains in completion rate while reducing end-to-end inference latency.

As summarized in Table 3, MM-ABC achieves significant performance improvements across all evaluation domains. Sequential pipelines fail frequently because real-world manipulation requires reactive whole-body compensation—the base must often micro-adjust its pose as the arm exerts force against refrigerator seals or drawer sliders. Monolithic flow models suffer from cross-stream mode mismatch, frequently hesitating during phase transitions. By combining decoupled multi-stream transformations with clean-action x -prediction, MM-ABC maintains high coordination accuracy, achieving a 74.5% success rate on RoboCasa-365 and 73.3% on physical robot trials with 112 ms fast inference.

6.3 Ablation Studies: Architectural Components

To isolate the contributions of MM-ABC’s individual modules, we perform systematic ablations on RoboCasa composite tasks:

- **World Expert Geometric Supervision:** Pruning the VGGT-Omega world query predictions (F_1, F_2) leads to a 28.4% drop in success rate during tasks involving moving base approaches. Without auxiliary 3D geometric rollout supervision, the policy struggles to compensate for visual parallax under ego-motion, frequently misjudging volumetric clearances.
- **DeepStack Multi-Layer Conditioning:** Removing intermediate DeepStack tokens and relying exclusively on the final VLM layer degrades task performance by 16.7%. Early and intermediate layers retain sharp high-frequency edge gradients essential for sub-centimeter contact alignment.
- **MM-JiT Structured Temporal Masking:** Replacing MM-JiT’s near-far attention masking with unconstrained full bidirectional attention across all horizons increases latency by 2.3× and introduces unstable arm jitter during navigation phases. The structured mask effectively isolates speculative far-horizon variance from immediate execution.

6.4 Practitioner Takeaways and Conclusion

Key Takeaways for Generalist Mobile Manipulation

1. **Architectural Decoupling with Shared Attention:** Mobile bases and manipulators possess fundamentally distinct dynamics, inertias, and frequencies. Independent parameterizations linked by masked joint attention prevent cross-stream interference without sacrificing coordination.
2. **Supervise Future Geometry without Deployment Overhead:** Metric spatial understanding is non-negotiable under ego-motion. Auxiliary future geometric objectives shape the visual latent space during training, requiring zero additional runtime cost.
3. **Clean-Action Prediction Enhances Partner Communication:** In finite-capacity multi-stream transformers, x -prediction avoids wasting representational bandwidth on high-dimensional flow noise, maximizing the capacity allocated to collaborative intent.

Conclusion. In this work, we introduced **MM-ABC**, an open foundation framework for scalable mobile manipulation. By uniting multi-level DeepStack perception (*Seeing*), structured dual-stream MM-JiT coordination with clean-action flow matching (*Coordinating*), and inference-free geometric world supervision (*Imagining*), MM-ABC addresses the fundamental spatial and dynamical bottlenecks of mobile embodied AI. Supported by our 5,000+ hour cross-embodiment pretraining corpus, MM-ABC establishes a robust and scalable paradigm for general-purpose robotic assistance in everyday human environments.

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